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Fish assemblages on shipwrecks and natural rocky reefs strongly differ in trophic structure

Thiony Simon*, Jean-Christophe Joyeux, Hudson T. Pinheiro

Departamento de Oceanografia e Ecologia, Universidade Federal do Espírito Santo, Av. Fernando Ferrari 514, Vitória, ES, 29075-910, Brazil

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A B S T R A C T

In the present work fish assemblages over two metallic vessels, five and 105 years old, and two natural rocky reefs were compared. The hypothesis that shipwrecks support assemblages with trophic structure similar to that encountered on natural reefs was rejected. Artificial and natural reefs strongly differ in their trophic structure, both in their multivariate composition and in biomass of most guilds. Substrate characteristics such as rugosity and benthic cover were found to influence the trophic organisation of the communities. Moreover, slow-paced structural changes over time in both biotic and abiotic aspects of wrecks appear responsible for younger and older artificial reefs be dissimilar in respect to biomass density of most feeding guilds. However, the older artificial reef did not present any strikingly "intermediate" feature between the younger artificial reef and the natural reefs, evidencing that distinct trophic assemblages exist over wrecks. Finally, the results found indicate that the use of shipwrecks as mitigation tool for losses of natural reefs may not be fully appropriate as they greatly differ in trophic structure, and consequently in energy flow, from natural reefs. Also, setting shipwrecks near natural reefs should be avoided as they differ in resources availability for many species, which may alter the community structure of natural habitats.

1. Introduction

One of the subjects most recurrently debated in the artificial reef literature is the attraction *vs.* production issue, i.e. whether artificial reefs produce new fish biomass or simply attract and aggregate fishes from natural reefs (e.g., Bohnsack et al., 1997; Bohnsack and Sutherland, 1985; Brickhill et al., 2005; Lindberg, 1997; Osenberg et al., 2002; Powers et al., 2003; Simon et al., 2011). Although this has frequently been treated as a dichotomic problem, attraction and production are only the extremes of a gradient that can change within and among species depending on the availability of natural reefs, mechanisms of natural population limitation, fishery exploitation pressure, life history dependence on reefs and species-specific and age-specific behavioural characteristics (Bohnsack, 1989).

Artificial reefs have been compared to natural ones in order to assess their performance, mainly when they are built aiming at compensating for habitat or resource loss due to human activities. In many cases, higher fish number and/or biomass density has been found at artificial reefs (e.g., Ambrose and Swarbrick, 1989; Arena et al., 2007; Bohnsack et al., 1994). However, such comparisons can be in some cases biased due to differences in reef characteristics. In fact, in the single study where reef size, age and isolation were controlled for both artificial and natural reefs (Carr and Hixon, 1997), the latter were found to support higher fish number and species than the former.

Benthic cover, rugosity, shelter availability, reef size and vertical relief are known to strongly influence fish assemblages (Charbonnel et al., 2002; Friedlander and Parrish, 1998; Gratwicke and Speight, 2005; Hixon and Brostoff, 1985; Kellison and Sedberry, 1998; McGehee, 1994). In particular, small artificial reefs show higher fish density while large ones support higher biomass density but fewer individuals (Bohnsack et al., 1994). However, the fact that multiple small reefs can support more individuals and species than one single reef of similar size (Bohnsack et al., 1994) indicates that ecotone development between reef and non-reef environments is important. The trophic structure of fish assemblages also appears to be determined by reef characteristics. For instance, the abundance of piscivores and planktivores can increase with depth while that of corallivores and mobile invertebrate feeders decreases (Friedlander and Parrish, 1998). Also, sessile invertebrate feeders can decrease, and piscivores and herbivores increase, with rugosity (Friedlander and Parrish, 1998). However,

* Corresponding author. Tel.: +55 2740097791; fax: +55 2740092500.
E-mail address: thionysimon@gmail.com (T. Simon).

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herbivores have been either positively (Floeter et al., 2007) or negatively (Friedlander and Parrish, 1998) related to algae cover. This apparent contradiction is due to the fact that, depending on locale, herbivores can be sustained by high algae availability or, inversely, the algal cover can be limited by high grazing pressure.

Many studies have compared fish communities between artificial and natural reefs (e.g., Randall, 1963; Rilov and Benayahu, 2000; Rooker et al., 1997; Stone et al., 1979; Terashima et al., 2007) and some have attempted to assess how much their trophic structures differ (e.g., Arena et al., 2007; Fowler and Booth, 2012; Hackradt et al., 2011; Honório et al., 2010). This approach provides a functional and ecological comparison rather than a taxonomic one (Friedlander and DeMartini, 2002; Friedlander and Parrish, 1998). The knowledge of the community trophic organization on both reef types allows assessing the effects that artificial reefs can apply over nearby natural reefs through habitat modifications (Bellwood et al., 2003, 2004; Mora et al., 2011). However, in many comparisons of artificial *vs*. natural reef fish communities, biotic and abiotic characteristics of the substrate have not been studied (e.g., Arena et al., 2007; Fowler and Booth, 2012; Hackradt et al., 2011), making it difficult to understand the association between reef and fauna. In the present work, the trophic structure of reef fish assemblages was compared among artificial and natural reefs in south-eastern Brazil to test the hypothesis that accidental or intentionally deployed vessels used as artificial reefs support fish assemblages with trophic structure similar to that of natural rocky reefs. Additionally, the implicit supposition that the trophic structure over artificial reefs changes as the artificial reef ages was examined. Finally, the biotic and abiotic reef characteristics that could be causing differences in trophic structure between artificial and natural reefs were investigated.

2. Material and methods

2.1. Study sites

Two pairs of artificial and natural reefs located about 10 km off Guarapari, south-eastern Brazil (Fig. 1) were examined. The two natural reefs, Escalvada and Rasas islands, are granitic and are located 5 km from each other. Escalvada is a single island and Rasas is composed by two small islands separated by a shallow and narrow strait. Depth at the interface between reef and unconsolidated substrate varies between 9 and 25 m depending on island side. The two artificial reefs, Bellucia and Victory, are steel-hulled freighters differing in origin and age. The 102-m Bellucia accidentally sunk in 1903 after colliding against a rock outcrop near Rasas Islands. In the collision, the ship broke in two parts that are now 150 m from each other. The maximum depth is 27 m and remains of the superstructure reach 20 m below the surface. The 90-m Victory was intentionally deployed in 2003 for disposal and to support tourism after being stripped of everything but paint. Its maximum depth is 35 m and the top of the funnel at the time of this study was 18 m below the surface. While the Bellucia is located less than 300 m from Rasas Islands, the Victory is about 2 km from Escalvada Island and 3 km from Rasas (Fig. 1). Note, however, that there also are many uncharted reef patches in the region with base of, either, granite or coralline algae and bryozoans or even reef-building corals. Both artificial reefs are located on extensive sandbanks, but parts of the Bellucia remain on the smooth, low-rugosity base of the rocky reef that sealed its fate.

2.2. Trophic structure of reef fish assemblages

Between January and March 2008, the reef fish assemblage was assessed by underwater visual census with 20×2 m strip transects. This period was chosen because only during the austral summer there is sufficient visibility (usually more than 5 m) to permit accurate visual observations. Each census was completed in two steps. In the first step, the diver randomly selected a starting point, swam unrolling a tape while counting the more mobile species that generally were larger-sized fish of demersal or pelagic habits. In the second step, the diver swam back to the initial point rolling up the tape and counting the more cryptic species that generally were smaller sized and of benthic habit. This visual census method is widely used on the Brazilian coast because it is suitable in low

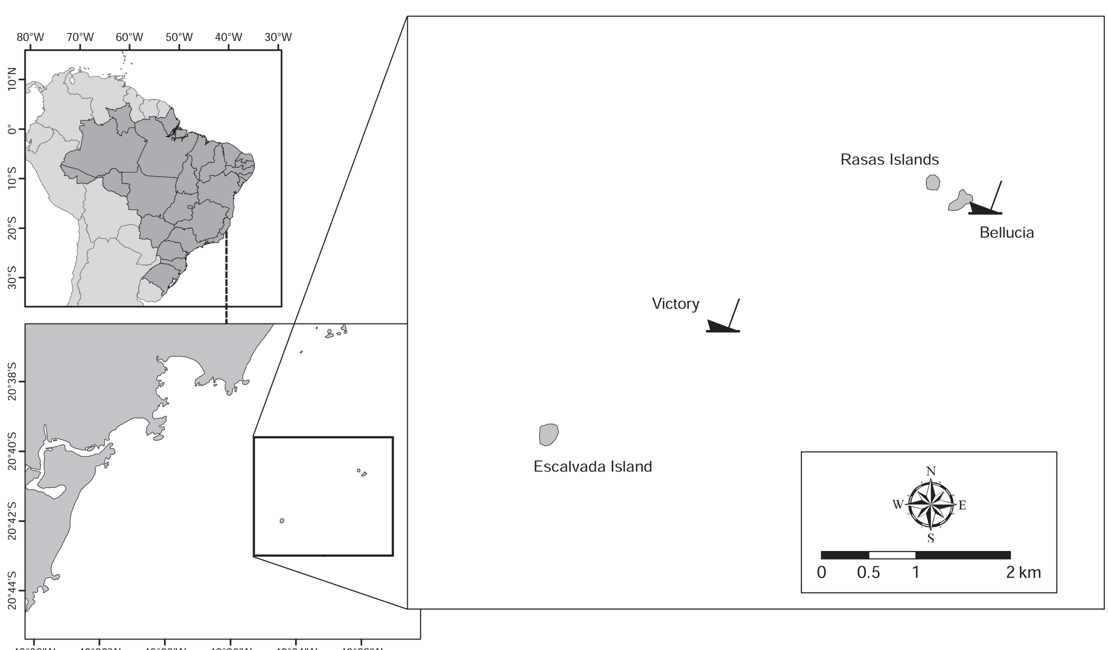

Fig. 1. The natural and artificial reefs studied, showing their insertion in the region of Guarapari, south-eastern Brazil.

visibility and covers a fixed area within a defined habitat. Also, density estimates for cryptobenthic species obtained using this method were deemed consistent by previous workers (Floeter et al., 2007; Krajewski and Floeter, 2011; Pinheiro et al., 2011). In each census, the number of individuals of each species was tallied along with their estimated total length (TL) in 10 cm size classes. An abundance scale was established to reduce the probability of error in enumerating individuals in schools. Individuals in schools up to 20 specimens were counted individually while larger schools were classified as containing 30, 50, 100, 200, 500, 1000 or 2000 fishes.

The number of censuses done on natural reefs (113 at Escalvada and 126 at Rasas) was higher than on artificial reefs (35 on the Bellucia and 46 on the Victory) due to the difference in total area between these sites (*authors pers. obs*). In order to sample almost all existing reef environments, transects were distributed into three or four sectors. Around islands, censuses were distributed along north, south, east and west sides. On shipwrecks, censuses were performed on stern, bow and superstructure. To avoid pseudoreplication, in each of these sectors censuses were further distributed into subsectors. Thus, in all island sectors censuses were stratified according to the depth gradient into three strata: surface (with 3 m as the shallowest sampling depth due to wave action), middle rocky shore (about halfway between surface and interface) and interface (the limit between hard and soft substrate). On the stern and bow of shipwrecks, censuses were performed on the main deck and cargo hold and at the interface (as in natural reefs). No further subdivision was applied for the superstructure of the Victory and censuses were performed on the horizontal surfaces defined by the upper decks. There are no significant remains of the Bellucia's superstructure.

All transects were performed by two trained divers using the standardised procedure described above. Training consisted in divers performing censuses while swimming side by side to ensure that the same assemblage was available to both. Differences in estimates for schools were debated after each dive until reaching a consensus. Subsequently to training, censuses were done concurrently by the two divers in the same sectors and subsectors.

Fish numbers and sizes were converted to biomass through length-weight equations (Froese and Pauly, 2008) using size centre-of-class. When no specific equation was available, an equation for a similar species or a mean genus, family or body shape equations was applied. Throughout the study, references to biomass actually refer to biomass density (in g m^{-2}). Species were grouped into eight trophic guilds following Ferreira et al. (2004), based on available literature for adult diet (e.g., Randall, 1967). In synthesis, roving herbivores (ROH) are generally large fishes that include detritus, turf algae and macroalgae in their diet; territorial herbivores (TEH) are small fishes that mainly consume turf algae farmed within vigorously defended territories; mobile invertebrate feeders (MIF) feed primarily on mobile invertebrates associated to both hard and soft bottoms; sessile invertebrate feeders (SIF) feed on hard substrate-associated sessile invertebrates such as sponges, cnidarians and ascidians; omnivores (OMN) feed on a mix of animal and plant material; planktivores (PLK) feed primarily on macro- and micro-zooplankton; carnivores (CAR) feed on both mobile invertebrates and fishes; and piscivores (PIS) feed mainly on fishes. Although feeding plasticity and ontogenetic shifts make allotting fish into independent feeding guilds difficult (Floeter et al., 2004), this approach is useful to assess the general patterns of trophic organisation and evaluate how biotic and abiotic characteristics of the habitat influence the community structure.

2.3. Substrate characteristics

The benthic composition was determined and the bottom rugosity measured to evaluate the influence of substrate on the trophic structure of the reef fish assemblages. Study sites were stratified into the sectors and subsectors previously defined. In the whole, 48 transects were performed at each natural reef and 15 on each artificial reef. Transects were 10 m long and data consisted of five photos and one measure of rugosity per transect. Rugosity was estimated through the chain-and-tape method (1 m long chain with 3.3 cm links). A rugosity index was calculated as the ratio of contoured *vs.* straight distance between the transect end points; the index value increases with superficial rugosity. This index has been demonstrated to be a good predictor of reef fish structure and composition (cf. Friedlander and Parrish, 1998; Rooker et al., 1997). Benthic composition was determined through photoquadrat analysis (photos showed an area of 27.2 $\times$ 20.4 cm) using the software CPCe v3.5 (Kohler and Gill, 2006). Twenty random points were distributed in each photoquadrat and the biotic or abiotic category at these points was registered. The biotic categories were defined to be taxonomically as broad as possible and to represent functional groups. However, lower taxonomic groups representing a prominent characteristic of the reefs (e.g., the octocoral *Carijoa riisei*) were maintained as individual categories. The biotic categories were "crustose coralline algae", "articulated coralline algae", "non-coralline algae", "stony corals", "firecorals", "anemones", "gorgonians", "*Carijoa riisei*", "hydroids", "bryozoans", "zoanthids", "sponges", "ascidians", "bivalves", "barnacles" and "crinoids". The abiotic categories were "sedimentation" (particulate or flocculate material deposited on the bottom), "unconsolidated substrate" (i.e., mud, sand or gravel deposits, indiscriminately) and "pavement" (bare rock or metal). The relative cover of each category was averaged for the five photoquadrats of the same transect.

2.4. Statistical analyses

Permutational Multivariate Analyses of Variance (PERMANOVA, $\alpha = 0.05$; Anderson et al., 2008) were performed to assess if the reef fish trophic structure differed between artificial and natural reefs (reef nature factor) and among reefs (site factor; with posterior pairwise comparisons). As depth is known to influence the distribution of reef fishes (Friedlander and Parrish, 1998) this was inserted in the two models as a fixed factor (5 m-depth classes; the few censuses made on artificial reefs at 20 m, one on Victory and three on Belucia, were pooled with the 21–25 m depth class). Type III sum of squares was used. The PERMANOVAs were applied using Bray–Curtis similarity matrix derived from the square-root transformed biomass of each trophic guild. The "permutation of residuals under a reduced model" method was chosen and 9999 permutations were performed.

Differences in biomass between artificial and natural reefs and among reefs were tested for each guild by Analyses of Variance (ANOVA, $\alpha = 0.05$; Zar, 2010), following the design previously described for PERMANOVAs. When significant differences between reef sites were detected, a Tukey post-hoc test was performed ($\alpha = 0.05$). Prior to running the ANOVAs, the relationship between the mean and the associated standard deviation for each combination of fixed factors was examined to ascertain which transformation of raw biomass data, if necessary, was required to approximate the test assumptions of normality and constant variance (Clarke and Warwick, 2001). For the guilds carnivores, mobile invertebrate feeders, omnivores, piscivores and planktivores was applied a logarithmic transformation, for roving and territorial herbivorous was applied a fourth-root transformation and for sessile invertebrate feeder was applied a square-root transformation.

Spatial relationships in trophic structure were explored through a cluster analysis coupled to a Similarity Profile permutation test (SIMPROF; Clarke and Gorley, 2006). The SIMPROF was used in order to assess if the clustered samples had a true multivariate pattern (i.e., are genuine clusters; $\alpha = 0.05$). The results were synthetized in a non-metric multi-dimensional scaling analysis (nMDS; Clarke and Gorley, 2006) where the genuine groups defined by SIMPROF were identified. These analyses were run over a Bray–Curtis similarity matrix derived from the square-root transformed data previously grouped for samples from the same site and depth class.

Similarity Percentage analysis (SIMPER; Clarke and Gorley, 2006) was applied to examine the contributions of benthic categories to the Bray–Curtis dissimilarity between reef types. The benthic categories individually contributing for at least 5% of the total dissimilarity were tested for differences in cover through Mann–Whitney *U* test ($\alpha = 0.05$; Zar, 2010). The influence of substrate characteristics (i.e., rugosity and benthic cover categories contributing >95% of total cover) over reef fish trophic structure was explored through a canonical correspondence analysis (ter Braak, 1986). Environmental data were standardized and community data were square-root transformed before examination. Data were grouped as in nMDS.

3. Results

3.1. Reef fish trophic structure

In total, 130 *taxa* were observed (see Appendix section for details), 114 on natural reefs (91 at Escalvada and 99 at Rasas) and 89 on artificial reefs (64 on the Victory and 68 on the Bellucia). The most speciose guilds were mobile invertebrate feeders and carnivores, with 40 and 27 *taxa*, respectively, and the least speciose were territorial herbivores and sessile invertebrate feeders, with four and seven *taxa*, respectively.

The biomass and frequency of occurrence of each guild is discriminated by reef nature and site in Table 1. Total biomass (all guilds combined) was more than three times greater on artificial (873.4 g m^{-2}) than on natural reefs (259.2 g m^{-2}). The dominant guilds on artificial reefs were mobile invertebrate feeders (616.2 g m^{-2}, 71% of total biomass) and omnivores (164.8 g m^{-2}, 19%) with the former present in all censuses while the latter was present in two-thirds only. About 90% of total biomass of these two guilds was from only one species, *Haemulon aurolineatum* in the first case and *Chaetodipterus faber* on the second. However, while *H. aurolineatum* was recorded in 90% of censuses only a few schools (5% of censuses) of large-sized *C. faber* were sighted. On natural reefs, dominant guilds were roving herbivores (72.8 g m^{-2}, 28%), omnivores (71.4 g m^{-2}, 28%) and mobile invertebrate feeders (57.6 g m^{-2}, 22%). Roving herbivores and mobile invertebrate feeders were present in almost all censuses while omnivores were recorded in less than a half. Of these guilds, only omnivores were dominated by a single species, *Diplodus argenteus*, which was responsible by 70% of the total biomass and was observed in more than a half of censuses.

On artificial reefs, total biomass was over four times higher on older Bellucia (1531.7 g m^{-2}) than on younger Victory (372.5 g m^{-2}). On the Bellucia the dominant guilds were mobile invertebrate feeders (1043.3 g m^{-2}, 68%) and omnivores (372.2 g m^{-2}, 24%). Both guilds were recorded in all censuses and were dominated by *H. aurolineatum* and *C. faber*, respectively. On the Victory, mobile invertebrate feeders, dominated by *H. aurolineatum*, was the guild with greatest biomass (291.2 g m^{-2}, 78%). It was followed by piscivores (38.3 g m^{-2}, 10%), dominated by *Caranx crysos* (about 70% of total guild biomass) and that was recorded in 35% of censuses. On natural reefs, total biomass was similar between reef sites (295.5 g m^{-2} on Escalvada and 226.7 g m^{-2} on Rasas). At Escalvada, the guilds with greatest biomasses were omnivores (100.0 g m^{-2}, 31%), roving herbivores (68.8 g m^{-2}, 23%) and mobile invertebrate feeders (66.9 g m^{-2}, 23%), while at Rasas they were roving herbivores (76.3 g m^{-2}, 34%), omnivores (53.9 g m^{-2}, 24%) and mobile invertebrate feeders (49.3 g m^{-2}, 22%). Guild constancy among censuses and guild-dominating species were as described above for natural reefs overall.

PERMANOVAs detected that reef fish trophic structure was significantly different between natural and artificial reefs (Pseudo-*F* = 19.9; *P*-perm < 0.001) and among the four reefs (Pseudo-*F* = 10.1; *P*-perm < 0.001). Depth was found to contribute significantly to the total sums of squares in both tests (test for reef nature, Pseudo-*F* = 8.1; test for reef sites, Pseudo-*F* = 5.2; all *P*-perm < 0.001). Pair-wise tests between reef sites detected differences for all comparisons (*P*-perm < 0.01). ANOVAs testing for differences between artificial and natural reefs (reef nature plus depth model; left side of Fig. 2) showed the biomass of all guilds to vary significantly with depth and the biomass of mobile invertebrate feeders, omnivores, planktivores and roving herbivores to vary according to reef type. The three former guilds had greater biomasses on artificial reefs and the latter on natural reefs. ANOVAs testing for differences between reefs (reef site plus depth model; right side of Fig. 2) showed the biomass of all guilds but territorial herbivores and sessile invertebrate feeders to differ between reefs and that of all guilds but piscivores and sessile invertebrate feeders to vary with depth. No difference was detected between the two natural reefs while artificial reefs differed from each other for all guilds but territorial herbivores and sessile invertebrate feeders (Tukey post-hoc tests; Fig. 2). Natural reefs had the highest roving herbivores and the lowest mobile invertebrate feeders biomass, the Bellucia the highest carnivores, planktivores and mobile invertebrate feeders biomass, and the Victory the highest piscivores and lowest roving herbivores and omnivores biomass (Fig. 2). For carnivores and piscivores, strong differences between the two artificial reefs (both guilds had their greatest biomass on one and their lowest on the other) precluded ANOVA to detect differences between artificial and natural reefs.

The nMDS analysis discriminate three assemblages at an arbitrary 60% similarity (Fig. 3). However, SIMPROF analysis (Fig. 3) performed over the cluster (not show) detected that one

Table 1
Biomass (mean ± S.D.) and frequency of occurrence (F_O) of reef fish trophic guilds on natural and artificial reefs in south-eastern Brazil.

Guild[a]	Artificial reefs					
	Victory ($n = 46$)		Bellucia ($n = 35$)		Total	
	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O
ROH	4.37 ± 14.1	0.33	22.4 ± 30.8	0.83	12.2 ± 24.4	0.54
TEH	0.01 ± 0.05	0.07	0.26 ± 0.42	0.43	0.12 ± 0.30	0.22
MIF	291 ± 411	1	1043 ± 1335	1	616 ± 997	1
SIF	5.50 ± 12.8	0.96	22.1 ± 31.3	1	12.7 ± 24.0	0.98
OMN	6.95 ± 14.4	0.35	372 ± 1106	1	165 ± 744	0.63
PLK	22.1 ± 141	0.63	48.2 ± 55.9	1	33.4 ± 113	0.79
CAR	4.16 ± 13.2	0.41	20.0 ± 38.1	0.63	11.0 ± 27.9	0.51
PIS	38.3 ± 70.0	0.72	3.26 ± 10.8	0.23	23.1 ± 55.8	0.51
Total	373 ± 468		1532 ± 1730		873 ± 1315	
Guild[a]	**Natural reefs**					
	Escalvada ($n = 113$)		Rasas ($n = 126$)		Total	
	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O
ROH	68.8 ± 71.3	1	76.3 ± 78.3	0.99	72.8 ± 75.0	1
TEH	0.53 ± 1.15	0.43	0.90 ± 2.22	0.51	0.73 ± 1.80	0.47
MIF	66.9 ± 121	0.97	49.3 ± 89.9	0.97	57.6 ± 106	0.97
SIF	13.5 ± 14.1	0.88	13.9 ± 17.4	0.88	13.7 ± 15.9	0.88
OMN	91.0 ± 117	0.44	53.9 ± 93.7	0.43	71.5 ± 107	0.44
PLK	12.3 ± 41.1	0.88	12.3 ± 35.1	0.84	12.3 ± 38.0	0.86
CAR	11.3 ± 58.4	0.44	13.6 ± 124	0.35	12.5 ± 98.2	0.39
PIS	31.2 ± 146	0.34	6.47 ± 56.1	0.16	18.2 ± 109	0.24
Total	295 ± 304		227 ± 226		259 ± 268	

[a] Guilds acronyms: ROV = roving herbivores, TEH = territorial herbivores, MIF = mobile invertebrate feeders, SIF = sessile invertebrate feeders, OMN = omnivores, PLK = planktivores, CAR = carnivores and PIS = piscivores.

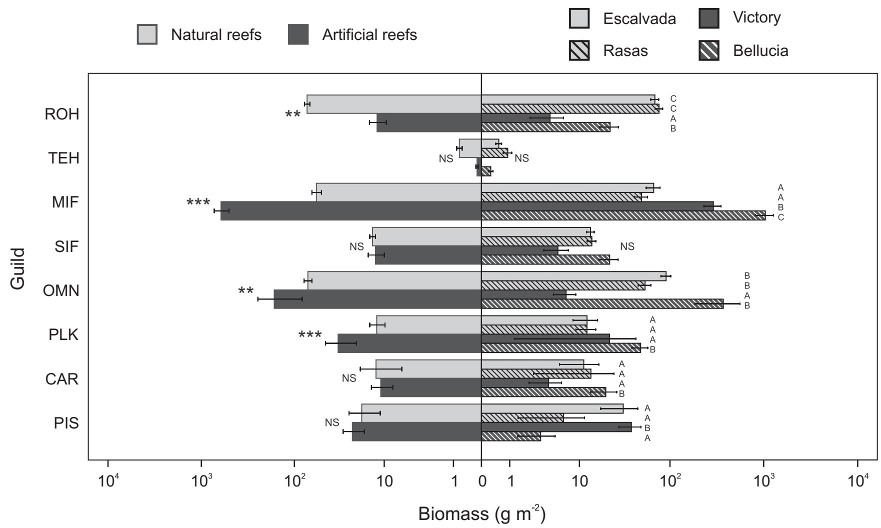

Fig. 2. Trophic structure (mean biomass ± S.E.) of reef fish assemblages on natural and artificial reefs in south-eastern Brazil. Guilds acronyms: ROV = roving herbivores, TEH = territorial herbivores, MIF = mobile invertebrate feeders, SIF = sessile invertebrate feeders, OMN = omnivores, PLK = planktivores, CAR = carnivores and PIS = piscivores. The results for the ANOVAs testing for differences between artificial and natural reefs are shown on the left side (***: $p \leq 0.001$; **: $p \leq 0.01$; NS: $p > 0.05$). Results of the Tukey post-hoc tests ($\alpha = 0.05$) for differences among sites (i.e., reefs) are shown on the right side: sites with identical letters are nor significantly different.

nMDS-assemblage was actually non-natural and a concatenation of two distinct groups. Then, group "a" is composed of all depth strata of artificial reefs but the deepest portion of the Victory (31–35 m). There, biomass of mobile invertebrate feeders (827.1 g m^{-2}), omnivores (199.8 g m^{-2}) and sessile invertebrate feeders (26.3 g m^{-2}) were the highest. Group "b" encompasses the deepest portion of Escalvada (21–26 m) plus the second deepest portion of Rasas (16–20 m); biomasses of carnivores (74.9 g m^{-2}) and piscivores (48.3 g m^{-2}) were the highest and of mobile invertebrate feeders (17.5 g m^{-2}) and planktivores (0.7 g m^{-2}) the lowest. Group "c" includes only the deepest portion of the Victory (31–35 m) where planktivores (30.4 g m^{-2}) reached its highest biomass and omnivores (4.9 g m^{-2}), roving herbivores (4.5 g m^{-2}), sessile invertebrate feeders (3.3 g m^{-2}) and territorial herbivores (<0.1 g m^{-2}) the lowest. Group "d" is composed of the remaining depth strata from both natural reefs and is characterized by the highest biomass of roving (73.3 g m^{-2}) and territorial herbivores (0.7 g m^{-2}) and the lowest of piscivores (12.7 g m^{-2}) and carnivores (4.8 g m^{-2}). The similarity between groups "c" and "d" in the nMDS was caused mainly by intermediate biomasses of mobile invertebrate feeders and planktivores and low biomass of carnivores.

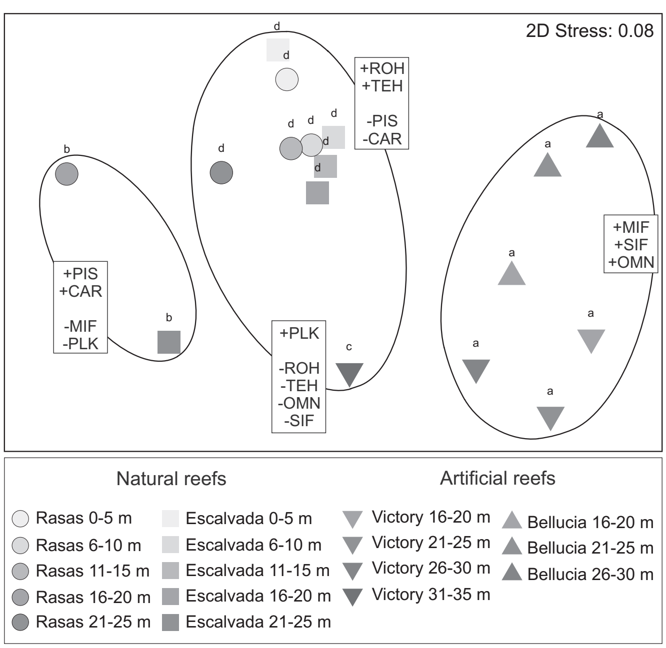

Fig. 3. Non-metric multidimensional scaling (nMDS) of the trophic structure of reef fish assemblages on natural and artificial reefs in south-eastern Brazil. Ellipses indicate 60% similarity. Groups (a, b, c, d) defined by similarity profile test are indicated as well the highest (+) and lowest (−) biomasses of each guild. Guilds acronyms as in Fig. 2.

3.2. Substrate characteristics

Rugosity was higher on artificial than on natural reefs, on the Bellucia than on the Victory and on Rasas than on Escalvada (the results of the Mann–Whitney *U* tests are presented in Table 2). On artificial reefs, particularly on the older Bellucia, pre-existent re-entrances into the structure (such as hatchways, door and hold openings) and external ironware (such as stairs and handrails) associated to corrosion and crushing of metallic walls made the surface highly rugged and irregular. Differences between the two natural reefs, despite statistically significant, were small.

Substrate categories differentiating artificial from natural reefs were "unconsolidated substrate" (19% contribution to total Bray–Curtis dissimilarity), "articulated coralline algae" (18%), "sedimentation" (13%), "bryozoans" (9%), "sponges" (9%), "crustose coralline algae" (8%), "non-coralline algae" (6%) and "*Carijoa riisei*" (5%). Algae covered a greater portion of natural (up to 45% of total cover, with articulated coralline algae reaching almost 24%) than of artificial reefs (5%; Table 2). Crustose coralline algae proportionally covered more area on the Bellucia than on the Victory. Bryozoans, *C. riisei* and sedimented material covered larger areas of artificial than natural reefs (Table 2). Although sponges cover was similar on artificial and natural reefs (20%), differences between the two artificial reefs (Table 2) probably caused the dissimilarity observed

Table 2
Rugosity index and benthic cover (%) of natural and artificial reefs in south-eastern Brazil. Benthic categories individually contributing for at least 5% of the total dissimilarity between artificial and natural reefs are boldfaced. Rugosity and contributing categories differing (Mann–Whitney *U* test; $\alpha = 0 \cdot 05$) between artificial and natural reefs (Φ), between artificial reefs (Ω) and between natural reefs (Ψ) are signalized.

	Artificial reefs (mean ± S.D.)			Natural reefs (mean ± S.D.)		
	Victory (n = 15)	Bellucia (n = 15)	Total	Escalvada (n = 45)	Rasas (n = 45)	Total
Rugosity index Φ Ω Ψ	1.32 ± 0.22	1.51 ± 0.22	1.41 ± 0.23	1.20 ± 0.11	1.22 ± 0.10	1.21 ± 0.11
Crustose coralline algae Φ Ω	**0.73 ± 1.58**	**3.75 ± 2.43**	**2.24 ± 2.53**	**9.19 ± 5.59**	**14.63 ± 14.01**	**11.91 ± 10.96**
Articulated coralline algae Φ	**0.34 ± 0.74**	**2.82 ± 2.52**	**1.58 ± 2.22**	**22.11 ± 17.42**	**25.65 ± 15.41**	**23.88 ± 16.46**
Non-coralline algae Φ	**1.81 ± 2.63**	**1.41 ± 1.47**	**1.61 ± 2.10**	**9.96 ± 10.57**	**7.58 ± 6.32**	**8.77 ± 8.74**
Stony corals	–	1.13 ± 2.50	0.57 ± 1.83	0.63 ± 1.62	0.17 ± 0.49	0.40 ± 1.21
Firecorals	–	–	–	0.65 ± 1.96	3.97 ± 8.79	2.31 ± 6.55
Anemones	0.07 ± 0.26	–	0.03 ± 0.18	–	–	–
Gorgonians	0.33 ± 0.82	–	0.17 ± 0.59	2.67 ± 4.69	1.45 ± 2.53	2.06 ± 3.80
***Carijoa riisei* Φ**	**4.68 ± 5.21**	**7.79 ± 6.32**	**6.23 ± 5.91**	**0.04 ± 0.20**	**0.08 ± 0.28**	**0.06 ± 0.25**
Hydroids	6.70 ± 4.71	4.63 ± 4.64	5.67 ± 4.71	1.83 ± 1.82	2.02 ± 2.33	1.92 ± 2.08
Bryozoans Φ Ψ	**16.37 ± 12.53**	**12.04 ± 5.89**	**14.20 ± 9.87**	**6.09 ± 7.55**	**2.95 ± 3.40**	**4.52 ± 6.03**
Zoanthids	0.07 ± 0.26	0.07 ± 0.26	0.07 ± 0.25	–	0.71 ± 3.33	0.35 ± 2.37
Sponges Ω	**15.28 ± 10.43**	**25.02 ± 8.84**	**20.15 ± 10.71**	**19.54 ± 8.87**	**21.06 ± 9.93**	**20.30 ± 9.40**
Ascidians	5.09 ± 5.54	0.07 ± 0.27	2.58 ± 4.62	0.98 ± 1.98	0.25 ± 0.64	0.61 ± 1.51
Bivalves	–	–	–	0.02 ± 0.14	0.02 ± 0.14	0.02 ± 0.14
Barnacles	0.07 ± 0.27	0.07 ± 0.26	0.07 ± 0.26	0.05 ± 0.23	–	0.02 ± 0.16
Crinoids	0.88 ± 1.51	–	0.44 ± 1.14	0.13 ± 0.64	0.23 ± 0.93	0.18 ± 0.80
Sedimentation Φ Ψ	**20.69 ± 23.15**	**15.76 ± 11.49**	**18.23 ± 18.13**	**5.39 ± 7.47**	**2.21 ± 4.72**	**3.80 ± 6.42**
Unconsolidated substrate	**25.29 ± 30.62**	**25.24 ± 18.86**	**25.26 ± 24.99**	**18.85 ± 20.84**	**15.19 ± 15.48**	**17.02 ± 18.35**
Pavement	1.33 ± 4.89	–	0.67 ± 3.47	0.02 ± 0.15	0.29 ± 1.24	0.16 ± 0.89

The boldfaced categories are those contributing for at least 5% in Bray–Curtis dissimilarity between artificial and natural reefs, as detected by SIMPER analysis.

between reef types. Overall, benthic cover was highly similar between natural reefs (Table 2).

3.3. Substrate influence on trophic structure

The two main axes in the CCA analysis (Fig. 4) explained 73% of the variance in the fish trophic structure (47% on Axis 1 and 26% on Axis 2). Herbivores, either roving or territorial, were associated with coralline algae (articulated or crustose) and shallow-water gorgonians and firecorals. Mobile invertebrate feeders are primarily correlated to unvegetated deeper areas where bryozoans, hydrozoans and sedimented-over substrates are dominant and where sponges, *Carijoa riisei* and rugose substrates are representative. Top predators such as piscivores and, especially, carnivores are indifferent to most substrate characteristics with a preference for unconsolidated bottoms. Sessile invertebrate feeders, planktivores and omnivores are, essentially, ubiquitous.

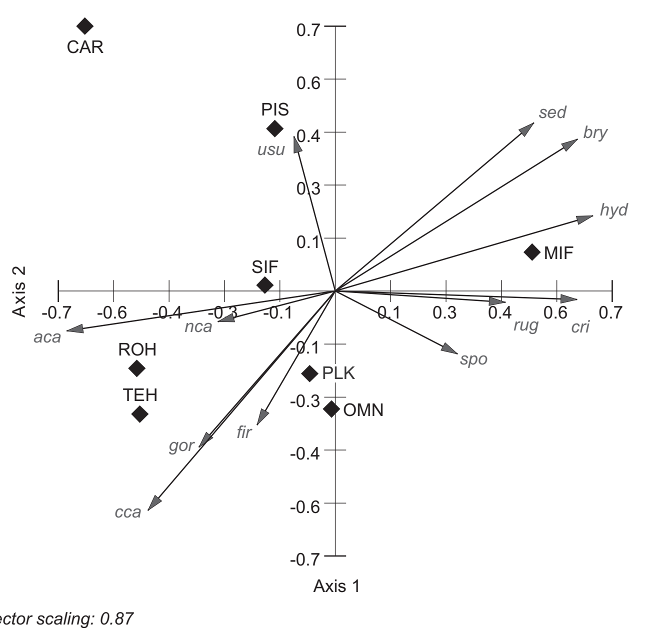

Fig. 4. Canonical correspondence analysis (CCA) evidencing the relationships between the trophic structure of reef fish assemblages and the bottom characteristics (substrate rugosity and benthic cover) of natural and artificial reefs in south-eastern Brazil. Environmental variables: articulate coralline algae (*aca*), crustose coralline algae (*cca*), non-coralline algae (*nca*), firecorals (*fir*), gorgonians (*gor*), sponges (*spo*), *Carijoa riisei* (*cri*), hydroids (*hyd*), bryozoans (*bry*), sedimentation (*sed*), unconsolidated substrate (*usu*) and rugosity (*rug*). Guilds acronyms as in Fig. 2.

4. Discussion

The main question addressed in the present work is if accidental or intentionally deployed vessels used as artificial reefs support fish assemblages with trophic structure similar to that of natural rocky reefs. Our results indicate that apparently this equivalence is not achieved, at least within the period considered, 100 years. The depth strata of artificial and natural reefs segregate in four distinct groups, one shallow and one deep over artificial reefs and two groups with weak depth affinities over natural reefs. The Bellucia, that we hypothesized to be more similar to natural reefs due to its older age and closer proximity to natural reefs, did not present any strikingly "intermediate" feature between the younger Victory and natural reefs. There is, thus, strong supporting evidence for distinct trophic assemblages over wrecks to exist and persist for extended periods of time, possibly much longer than a century. Slow-paced structural changes over time in both biotic (i.e., benthic cover) and abiotic (e.g., rugosity and relief) aspects of wrecks appear responsible for younger and older artificial reefs be dissimilar in respect to biomass density of most feeding guilds. Both the present work and a study done in north-eastern Brazil (Honório et al., 2010) indicate that close similarity in trophic structure between century-old shipwrecks and adjacent natural reefs, supposedly resulting from the effect of ageing on structure stabilisation (such as promoted by structure corrosion and crumbling) and colonisation (by either benthos or fish), should not be expected [but see Fowler and Booth (2012) for a counter-example].

In the present study, mobile invertebrate feeders accounted for the greatest difference in the trophic structure of fish assemblages over natural and artificial reefs. *H. aurolineatum* drove this pattern because, although being the species with the highest biomass on both reef types, its biomass was forty times higher on artificial than on natural reefs. Its density was greater on areas dominated by bryozoans, hydrozoans, sponges and *Carijoa riisei* and where the substrate is very rugose and partially covered by a layer of sediment. Other studies have found great differences in the density of this or other species from the same genus between artificial and natural reefs (Arena et al., 2007; Bohnsack et al., 1994; Hackradt et al., 2011; Honório et al., 2010). There also is some evidence that *H. aurolineatum* recruited more onto artificial than over natural reefs during the present study (Simon et al., 2011), but this is not sufficient to explain the strong differences observed. As other grunts, *H. aurolineatum* is a reef-associated species that rests over hard bottoms during daytime and migrates to adjacent soft bottoms at night, where it forages mainly on infaunal invertebrates (Burke, 1995; Nagelkerken et al., 2000; Ogden and Ehrlich, 1977; Sedberry, 1985). The area suitable for *H. aurolineatum* to rest is much smaller on artificial than natural reefs. Considering that foraging distance from the reef edge and density in the foraging area are independent from reef nature, the ratio between soft-foraging and hard-resting areas would be much higher for artificial reefs, and individuals would be at greater density there. On the other hand, *H. aurolineatum* biomass was more than three times higher on the Bellucia than on the Victory. Top-down effects driven by the removal of *H. aurolineatum* biomass by predation are possibly higher on the Victory, because piscivores biomass was almost 12-times higher there than on the Bellucia.

H. aurolineatum can also take advantage of planktonic food (Randall, 1967; Sedberry, 1985) during the day (Arena et al., 2007; pers. obs.). This is a plausible explanation if taking the higher biomass of planktivores on the Bellucia as a suggestion that plankton is more readily available or plentiful there, and taking into account that on the Victory *H. aurolineatum* biomass was much higher on the superstructure than near the deck and bottom interface. At highest locales on the Victory, biomass reached a value similar to that observed on the Bellucia, suggesting that these fishes aggregate were plankton is more abundant due to current and wave surge (Hamner et al., 1988; Thresher, 1983).

Fishes feeding on plankton are strongly influenced by wave motion, current strengths and, as many species are selective visual predators, water transparency (Bray, 1981; Clarke et al., 2009; Hamner et al., 1988; Lazzaro, 1987; Thresher, 1983). Together, these factors determine both food availability and feeding performance. The studied natural reefs, that have a significant portion reaching to (and above) the surface, may have lower planktivores biomass due to the fact that these physical barriers create an important resistance to water movements, diminishing currents speed in their periphery and, consequently, plankton flow. In contrast, both artificial reefs have the upper water column free of obstacle as the shipwrecks reach up to 18 or 20 m below the surface. Other works have found higher biomass of planktivores on artificial than on natural reefs (Arena et al., 2007; Rilov and Benayahu, 2000), indicating that these man-made structures can offer a higher availability of food resources than nearby natural settings.

A number of depth-related differences among reefs were expected because both artificial reefs are deeper than natural reefs and the remains of the Bellucia are overall closer to surface than the Victory. The guilds most affected would essentially be herbivorous, as algae cover and productivity tend to diminish as depth increases and light availability declines. In fact, both roving herbivores and territorial herbivores were significantly different between depth classes and their biomasses were greater where algae covered a greater portion of substrate. However, territorial herbivores were solely influenced by depth while roving herbivores were also influenced by reef site (biomass was much higher on natural than on artificial reefs and on the Bellucia than on the Victory). Preference for an algal diet would also explain the higher biomass of the omnivores species sergeant-major *Abudefduf saxatilis* and South American silver porgy *D. argenteus* on natural reefs. However, the influence of specific needs determinant to their feeding or reproductive success, such as shallow depth or adequate substrate for nests (e.g., *A. saxatilis*), cannot be ruled out.

While differences between selected genera of carnivores and piscivores targeted by local fisheries have been detected between artificial and natural reefs (Simon et al., 2011), these two guilds were not significantly different between reef types, only between reef sites. The biomass of carnivores was higher on the Bellucia and that of piscivores was higher on the Victory than on remaining site. Most carnivores biomass was represented by the reef croaker *Odontoscion dentex* and the dog snapper *Lutjanus jocu* on the Bellucia. Higher piscivores biomass on the Victory was caused mainly by the blue runner *Caranx crysos*, the comb grouper *Mycteroperca acutirostris* and the yellowmouth grouper *Mycteroperca interstitialis*. The structural difference between Bellucia (more crushed with smaller crevices and re-entrances) and Victory (well preserved with larger and protected empty spaces) can actually determine abundance patterns for predatory fish. In fact, fish density and size are directly correlated to recesses quantity and size (Hixon and Beets, 1989, 1993) and these are expected to varying according with age, thus strongly influencing the trophic structure of artificial reefs.

5. Conclusions

This study is among the first rigorous comparisons of the trophic structure of fish assemblages between artificial and natural reefs. Our results show that (1) large differences between reef nature may persist over decades and the distance between artificial reefs to natural reefs does not exert decisive influence in the similarity among them, (2) the divergences are related with difference in benthic cover and substrate features which have a close relationship with the reef fish community, as well as resultant ecological dynamics and (3) the evolution of artificial reef matter (due increase of rugosity), but not make its reef fish community more similar to natural ones.

Rocky reefs and steel shipwrecks are therefore intrinsically different as they support, and keep supporting, distinct fish assemblages. This implicates that, at least as a mitigation tool for losses due to human activities on natural reefs (Rilov and Benayahu, 1998, 2000), the use of derelict vessels, and probably other metallic obsolete materials, as artificial reefs is not adequate. Moreover, the establishment of shipwrecks near natural rocky reefs should be avoided, as its offers habitats differing in physical structure and thus can be attractive to some species, which can migrate from natural to artificial reefs, altering the community structure (Simon et al., 2011).

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Appendix

Appendix I

Biomass (mean ± S.D.) and frequency of occurrence (F_O) of reef fish species on natural and artificial reefs in south-eastern Brazil.

Species	Guild[a]	Artificial reefs						Natural reefs					
		Victory ($n = 46$)		Bellucia ($n = 35$)		Total		Escalvada ($n = 113$)		Rasas ($n = 126$)		Total	
		Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O
Acanthurus bahianus	ROH	0.42 ± 1.60	0.09	2.72 ± 4.54	0.54	1.41 ± 3.39	0.28	30.18 ± 26.32	0.98	28.37 ± 35.17	0.96	29.22 ± 31.25	0.97
Acanthurus chirurgus	ROH	1.94 ± 7.48	0.11	10.77 ± 14.22	0.46	5.75 ± 11.69	0.26	18.32 ± 28.14	0.5	21.55 ± 43.15	0.52	20.02 ± 36.78	0.51
Acanthurus coeruleus	ROH	1.96 ± 7.98	0.07	–	–	1.11 ± 6.06	0.04	0.40 ± 4.24	0.01	0.54 ± 4.54	0.02	0.47 ± 4.39	0.02
Cryptotomus roseus	ROH	–	–	–	–	–	–	0.37 ± 2.71	0.03	0.26 ± 1.53	0.05	0.31 ± 2.16	0.04
Kyphosus spp.	ROH	–	–	–	–	–	–	10.98 ± 50.80	0.09	2.81 ± 20.95	0.04	6.67 ± 38.23	0.06
Scarus trispinosus	ROH	–	–	0.67 ± 3.96	0.03	0.29 ± 2.60	0.01	–	–	–	–	–	–
Scarus zelindae	ROH	–	–	0.67 ± 3.96	0.03	0.29 ± 2.60	0.01	–	–	–	–	–	–
Sparisoma amplum	ROH	–	–	–	–	–	–	0.17 ± 1.79	0.02	0.01 ± 0.13	0.01	0.09 ± 1.23	0.01
Sparisoma axillare	ROH	0.01 ± 0.06	0.04	2.71 ± 11.42	0.06	1.18 ± 7.56	0.05	7.75 ± 27.76	0.24	18.74 ± 40.85	0.35	13.54 ± 35.63	0.30
Sparisoma frondosum	ROH	0.03 ± 0.22	0.02	4.79 ± 12.21	0.20	2.09 ± 8.31	0.10	0.59 ± 3.21	0.04	4.06 ± 12.92	0.15	2.42 ± 9.78	0.10
Sparisoma radians	ROH	–	–	–	–	–	–	–	–	0.01 ± 0.13	0.01	0.01 ± 0.09	<0.01
Sparisoma spp. (juv)	ROH	0.19 ± 0.03	0.04	0.01 ± <0.01	0.03	0.19 ± 0.02	0.04	1.45 ± 0.08	0.14	0.29 ± 0.01	0.08	1.74 ± 0.05	0.11
Sparisoma tuiupiranga	ROH	–	–	0.04 ± 0.25	0.03	0.02 ± 0.16	0.01	0.07 ± 0.36	0.05	–	–	0.03 ± 0.25	0.03
Microspathodon chrysurus	TEH	–	–	–	–	–	–	–	–	0.02 ± 0.22	0.01	0.01 ± 0.16	<0.01
Stegastes fuscus	TEH	–	–	–	–	–	–	0.23 ± 0.96	0.16	0.58 ± 2.05	0.19	0.41 ± 1.63	0.18
Stegastes pictus	TEH	0.01 ± 0.05	0.02	0.26 ± 0.42	0.43	0.12 ± 0.30	0.20	0.28 ± 0.67	0.30	0.27 ± 0.57	0.30	0.27 ± 0.62	0.30
Stegastes variabilis	TEH	<0.01 ± <0.01	0.04	–	–	<0.01 ± <0.01	0.02	0.02 ± 0.09	0.09	0.04 ± 0.12	0.18	0.03 ± 0.11	0.14
Amblycirrhitus pinos	MIF	–	–	0.02 ± 0.04	0.17	0.01 ± 0.03	0.07	0.04 ± 0.14	0.14	0.19 ± 0.78	0.22	0.12 ± 0.58	0.18
Anisotremus moricandi	MIF	–	–	1.97 ± 7.56	0.14	0.85 ± 5.03	0.06	0.90 ± 3.50	0.09	0.63 ± 2.62	0.07	0.76 ± 3.06	0.08
Anisotremus surinamensis	MIF	2.52 ± 7.90	0.11	–	–	1.43 ± 6.06	0.06	1.33 ± 10.17	0.04	0.04 ± 0.45	0.01	0.65 ± 7.01	0.02
Anisotremus virginicus	MIF	4.25 ± 12.51	0.33	4.72 ± 9.52	0.26	4.45 ± 11.25	0.30	2.99 ± 11.34	0.22	7.41 ± 25.08	0.26	5.32 ± 19.89	0.24
Balistes vetula	MIF	–	–	–	–	–	–	–	–	0.12 ± 0.70	0.04	0.06 ± 0.51	0.02
Bodianus pulchellus	MIF	1.77 ± 3.63	0.37	3.65 ± 9.18	0.37	2.58 ± 6.64	0.37	0.07 ± 0.74	0.03	0.03 ± 0.21	0.02	0.05 ± 0.53	0.03
Bodianus rufus	MIF	2.96 ± 7.56	0.26	3.76 ± 7.74	0.51	3.31 ± 7.60	0.37	7.84 ± 17.08	0.46	8.83 ± 14.95	0.50	8.36 ± 15.97	0.48
Calamus spp.	MIF	–	–	–	–	–	–	0.79 ± 6.00	0.02	0.16 ± 1.60	0.02	0.46 ± 4.28	0.02
Callionymus bairdi	MIF	–	–	–	–	–	–	–	–	<0.01 ± <0.01	0.02	<0.01 ± <0.01	0.01
Coryphopterus dicrus	MIF	0.03 ± 0.10	0.35	0.01 ± 0.03	0.46	0.02 ± 0.08	0.40	0.01 ± 0.02	0.24	<0.01 ± 0.01	0.16	<0.01 ± 0.01	0.20
Coryphopterus glaucofraenum	MIF	<0.01 ± <0.01	0.02	–	–	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.03	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.02
Dactylopterus volitans	MIF	–	–	–	–	–	–	1.92 ± 20.36	0.01	–	–	0.91 ± 14.00	<0.01
Diodon hystrix	MIF	1.29 ± 6.14	0.04	3.30 ± 15.22	0.06	2.16 ± 10.99	0.05	1.58 ± 8.75	0.04	0.90 ± 7.11	0.02	1.22 ± 7.92	0.03
Doratonotus megalepis	MIF	–	–	–	–	–	–	–	–	<0.01 ± <0.01	0.02	<0.01 ± <0.01	0.01
Elacatinus figaro	MIF	–	–	<0.01 ± <0.01	0.06	<0.01 ± <0.01	0.02	<0.01 ± <0.01	0.06	<0.01 ± <0.01	0.02	<0.01 ± <0.01	0.04
Emblemariopsis signifer	MIF	–	–	–	–	–	–	–	–	<0.01 ± <0.01	0.06	<0.01 ± <0.01	0.03
Equetus lanceolatus	MIF	–	–	–	–	–	–	–	–	<0.01 ± <0.01	0.01	<0.01 ± <0.01	<0.01
Eucinostomus argenteus	MIF	–	–	–	–	–	–	–	–	0.01 ± 0.10	0.01	<0.01 ± 0.07	<0.01
Gramma brasiliensis	MIF	<0.01 ± 0.01	0.11	0.01 ± 0.03	0.40	0.01 ± 0.02	0.23	0.01 ± 0.07	0.19	<0.01 ± 0.02	0.12	0.01 ± 0.05	0.15
Haemulon aurolineatum	MIF	273.22 ± 408.29	0.85	974.00 ± 1337.07	0.94	576.03 ± 987.72	0.89	23.14 ± 98.15	0.14	5.74 ± 31.98	0.16	13.97 ± 71.74	0.15
Haemulon parra	MIF	–	–	2.59 ± 9.52	0.11	1.12 ± 6.34	0.05	–	–	0.05 ± 0.57	0.01	0.03 ± 0.41	<0.01
Haemulon plumierii	MIF	–	–	17.00 ± 24.99	0.69	7.35 ± 18.36	0.30	8.39 ± 24.69	0.37	10.46 ± 49.48	0.18	9.48 ± 39.67	0.27
Haemulon steindachneri	MIF	1.97 ± 5.74	0.22	12.51 ± 64.02	0.20	6.52 ± 42.28	0.21	5.32 ± 37.07	0.05	1.70 ± 9.17	0.08	3.41 ± 26.35	0.07
Halichoeres brasiliensis	MIF	0.41 ± 1.28	0.48	0.55 ± 0.94	0.80	0.47 ± 1.14	0.62	0.52 ± 2.42	0.31	0.71 ± 2.52	0.39	0.62 ± 2.47	0.35
Halichoeres dimidiatus	MIF	–	–	0.18 ± 0.55	0.11	0.08 ± 0.37	0.05	–	–	0.53 ± 2.07	0.15	0.28 ± 1.52	0.08
Halichoeres penrosei	MIF	0.28 ± 1.88	0.02	–	–	0.16 ± 1.42	0.01	0.02 ± 0.18	0.04	0.30 ± 1.29	0.11	0.17 ± 0.96	0.08
Halichoeres poeyi	MIF	0.78 ± 1.05	0.76	0.90 ± 1.00	0.80	0.83 ± 1.02	0.78	0.70 ± 1.42	0.51	1.30 ± 2.31	0.63	1.02 ± 1.96	0.57
Halichoeres sazimai	MIF	0.12 ± 0.47	0.07	–	–	0.07 ± 0.36	0.04	–	–	–	–	–	–
Holocentrus adscensionis	MIF	1.24 ± 3.16	0.30	14.39 ± 22.35	0.74	6.92 ± 16.15	0.49	7.43 ± 15.05	0.44	8.39 ± 27.74	0.37	7.93 ± 22.60	0.41
Hypleurochilus spp.	MIF	–	–	<0.01 ± <0.01	0.03	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.01	<0.01 ± 0.01	0.03	<0.01 ± 0.01	0.02
Malacoctenus aff. *triangulatus*	MIF	0.01 ± 0.02	0.22	0.01 ± 0.01	0.60	0.01 ± 0.02	0.38	0.01 ± 0.01	0.69	0.01 ± 0.02	0.75	0.01 ± 0.01	0.72
Malacoctenus delalandii	MIF	<0.01 ± <0.01	0.02	–	–	<0.01 ± <0.01	0.01	<0.01 ± 0.01	0.07	<0.01 ± <0.01	0.10	<0.01 ± 0.01	0.08
Mulloidichthys martinicus	MIF	0.11 ± 0.75	0.02	0.70 ± 2.71	0.17	0.36 ± 1.88	0.09	0.38 ± 2.90	0.04	0.11 ± 0.91	0.02	0.24 ± 2.10	0.03

Opistognathus whitehursti	MIF	–	–	–	–	–	–	<0.01 ± <0.01	0.01	–	–	<0.01 ± <0.01	<0.01
Pareques acuminatus	MIF	0.03 ± 0.19	0.04	2.65 ± 6.07	0.43	1.16 ± 4.17	0.21	0.46 ± 2.19	0.09	0.04 ± 0.29	0.06	0.24 ± 1.53	0.08
Pseudupeneus maculatus	MIF	0.03 ± 0.20	0.02	0.26 ± 1.09	0.09	0.13 ± 0.73	0.05	2.99 ± 7.76	0.27	1.53 ± 5.85	0.17	2.22 ± 6.85	0.22
Serranus atrobranchus	MIF	0.01 ± 0.04	0.15	–	–	0.01 ± 0.03	0.09	–	–	–	–	–	–
Serranus baldwini	MIF	0.01 ± 0.06	0.20	0.10 ± 0.13	0.60	0.05 ± 0.11	0.37	0.05 ± 0.13	0.26	0.08 ± 0.18	0.27	0.06 ± 0.16	0.26
Serranus flaviventris	MIF	0.18 ± 0.26	0.76	0.01 ± 0.04	0.06	0.11 ± 0.21	0.46	<0.01 ± <0.01	0.02	–	–	<0.01 ± <0.01	0.01
Sphoeroides spengleri	MIF	–	–	–	–	–	–	0.02 ± 0.18	0.03	0.02 ± 0.07	0.05	0.02 ± 0.13	0.04
Canthigaster figueiredoi	SIF	0.10 ± 0.19	0.37	0.30 ± 0.50	0.60	0.19 ± 0.37	0.47	0.22 ± 0.58	0.42	0.29 ± 0.68	0.50	0.26 ± 0.63	0.46
Chaetodon sedentarius	SIF	0.66 ± 1.00	0.91	0.65 ± 1.84	0.63	0.65 ± 1.41	0.79	0.18 ± 0.88	0.19	0.05 ± 0.26	0.10	0.11 ± 0.64	0.14
Chaetodon striatus	SIF	0.01 ± 0.04	0.07	1.09 ± 2.12	0.23	0.48 ± 1.48	0.14	2.25 ± 3.06	0.45	1.62 ± 2.79	0.33	1.92 ± 2.93	0.39
Chilomycterus reticulatus	SIF	–	–	2.46 ± 14.57	0.03	1.06 ± 9.58	0.01	–	–	–	–	–	–
Chilomycterus spinosus	SIF	0.30 ± 2.03	0.02	–	–	0.17 ± 1.53	0.01	–	–	0.64 ± 4.06	0.03	0.34 ± 2.96	0.02
Holacanthus ciliaris	SIF	4.39 ± 12.74	0.20	6.46 ± 15.52	0.26	5.28 ± 13.95	0.22	1.60 ± 6.17	0.08	2.82 ± 7.80	0.19	2.24 ± 7.09	0.14
Holacanthus tricolor	SIF	0.05 ± 0.36	0.02	11.13 ± 22.62	0.46	4.84 ± 15.75	0.21	9.22 ± 11.63	0.62	8.49 ± 14.22	0.52	8.84 ± 13.04	0.57
Abudefduf saxatilis	OMN	–	–	0.11 ± 0.42	0.09	0.05 ± 0.28	0.04	15.34 ± 67.09	0.37	5.01 ± 10.36	0.32	9.89 ± 46.92	0.34
Acanthostracion polygonius	OMN	0.15 ± 1.03	0.02	0.55 ± 3.24	0.03	0.32 ± 2.26	0.02	0.55 ± 2.76	0.06	0.60 ± 3.05	0.06	0.58 ± 2.91	0.06
Acanthostracion quadricornis	OMN	0.52 ± 3.55	0.02	–	–	0.30 ± 2.68	0.01	0.07 ± 0.63	0.02	0.02 ± 0.18	0.01	0.04 ± 0.45	0.01
Cantherhines macrocerus	OMN	–	–	–	–	–	–	1.59 ± 6.39	0.07	0.51 ± 2.51	0.05	1.02 ± 4.78	0.06
Cantherhines pullus	OMN	0.12 ± 0.78	0.02	1.37 ± 3.36	0.23	0.66 ± 2.35	0.11	1.01 ± 2.50	0.21	1.47 ± 3.07	0.30	1.25 ± 2.82	0.26
Centropyge aurantonotus	OMN	–	–	–	–	–	–	–	–	<0.01 ± <0.01	0.01	<0.01 ± <0.01	<0.01
Chaetodipterus faber	OMN	–	–	335.50 ± 1090.51	0.11	144.97 ± 730.33	0.05	–	–	–	–	–	–
Diplodus argenteus	OMN	1.09 ± 4.48	0.07	24.70 ± 45.13	0.69	11.29 ± 31.86	0.33	64.60 ± 93.18	0.65	36.92 ± 89.58	0.44	50.01 ± 92.15	0.54
Gnatholepis thompsoni	OMN	<0.01 ± <0.01	0.02	–	–	<0.01 ± <0.01	0.01	–	–	–	–	–	–
Hypsoblennius invemar	OMN	–	–	–	–	–	–	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.01
Parablennius marmoreus	OMN	0.03 ± 0.05	0.54	0.01 ± 0.03	0.86	0.02 ± 0.04	0.68	0.01 ± 0.01	0.52	0.01 ± 0.02	0.48	0.01 ± 0.01	0.50
Parablennius pilicornis	OMN	–	–	–	–	–	–	<0.01 ± <0.01	0.01	<0.01 ± 0.01	0.01	<0.01 ± 0.01	0.01
Pomacanthus arcuatus	OMN	4.30 ± 13.22	0.11	6.70 ± 25.31	0.09	5.34 ± 19.29	0.10	1.17 ± 6.13	0.04	2.87 ± 11.95	0.08	2.06 ± 9.66	0.06
Pomacanthus paru	OMN	0.74 ± 5.02	0.02	3.26 ± 9.78	0.11	1.83 ± 7.51	0.06	6.64 ± 26.52	0.11	6.52 ± 18.50	0.15	6.58 ± 22.60	0.13
Stephanolepis hispidus	OMN	–	–	–	–	–	–	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.01	<0.01 ± <0.01	0.01
Apogon americanus	PLK	–	–	–	–	–	–	<0.01 ± 0.02	0.01	–	–	<0.01 ± 0.01	<0.01
Chromis flavicauda	PLK	–	–	–	–	–	–	<0.01 ± 0.02	0.01	0.02 ± 0.14	0.03	0.01 ± 0.10	0.02
Chromis jubauna	PLK	<0.01 ± <0.01	0.02	0.03 ± 0.11	0.09	0.01 ± 0.07	0.05	–	–	–	–	–	–
Chromis multilineata	PLK	0.15 ± 0.71	0.11	0.80 ± 1.46	0.40	0.43 ± 1.14	0.23	6.72 ± 34.40	0.29	1.81 ± 8.05	0.24	4.13 ± 24.43	0.26
Clepticus brasiliensis	PLK	–	–	19.12 ± 50.05	0.34	8.26 ± 33.99	0.15	1.57 ± 9.52	0.06	4.18 ± 17.11	0.13	2.95 ± 14.08	0.10
Clupeidae	PLK	20.89 ± 141.25	0.04	–	–	11.86 ± 106.45	0.02	–	–	–	–	–	–
Decapterus spp.	PLK	0.96 ± 2.97	0.11	0.32 ± 1.89	0.03	0.68 ± 2.57	0.07	–	–	–	–	–	–
Myripristis jacobus	PLK	–	–	23.57 ± 35.66	0.60	10.18 ± 26.05	0.26	3.51 ± 11.37	0.13	1.67 ± 7.70	0.08	2.54 ± 9.63	0.10
Opistognathus aff. *aurifrons*	PLK	–	–	0.01 ± 0.07	0.06	0.01 ± 0.04	0.02	0.01 ± 0.09	0.04	<0.01 ± 0.01	0.01	0.01 ± 0.07	0.02
Paranthias furcifer	PLK	<0.01 ± 0.02	0.02	3.43 ± 6.75	0.31	1.48 ± 4.72	0.15	0.36 ± 2.56	0.04	4.19 ± 21.92	0.08	2.38 ± 16.09	0.06
Pempheris schomburgkii	PLK	–	–	0.92 ± 3.06	0.09	0.40 ± 2.05	0.04	0.01 ± 0.11	0.01	–	–	0.01 ± 0.08	<0.01
Pseudocaranx dentex	PLK	–	–	–	–	–	–	–	–	0.03 ± 0.38	0.01	0.02 ± 0.27	<0.01
Ptereleotris randalli	PLK	0.05 ± 0.34	0.07	–	–	0.03 ± 0.26	0.04	0.10 ± 0.66	0.05	0.02 ± 0.13	0.02	0.06 ± 0.46	0.04
Thalassoma noronhanum	PLK	–	–	–	–	–	–	–	–	0.32 ± 1.22	0.08	0.17 ± 0.90	0.04
Alphestes afer	CAR	–	–	–	–	–	–	–	–	0.13 ± 1.48	0.01	0.07 ± 1.08	<0.01
Bothus lunatus	CAR	–	–	–	–	–	–	–	–	0.01 ± 0.12	0.01	0.01 ± 0.09	<0.01
Bothus ocellatus	CAR	–	–	–	–	–	–	0.01 ± 0.11	0.01	–	–	<0.01 ± 0.07	<0.01
Cephalopholis fulva	CAR	0.02 ± 0.06	0.15	1.20 ± 3.81	0.17	0.53 ± 2.55	0.16	1.34 ± 6.16	0.13	1.31 ± 5.03	0.14	1.32 ± 5.58	0.14
Dasyatis centroura	CAR	–	–	4.59 ± 27.17	0.03	1.98 ± 17.86	0.01	–	–	11.03 ± 123.86	0.01	5.82 ± 89.93	<0.01
Diplectrum radiale	CAR	0.12 ± 0.61	0.11	–	–	0.07 ± 0.46	0.06	–	–	–	–	–	–
Elops saurus	CAR	–	–	–	–	–	–	5.33 ± 56.63	0.01	–	–	2.52 ± 38.94	<0.01
Fistularia tabacaria	CAR	–	–	–	–	–	–	0.60 ± 4.27	0.03	–	–	0.29 ± 2.94	0.01
Gymnothorax funebris	CAR	0.04 ± 0.26	0.02	–	–	0.02 ± 0.20	0.01	0.02 ± 0.17	0.01	–	–	0.01 ± 0.11	<0.01
Gymnothorax miliaris	CAR	–	–	–	–	–	–	–	–	<0.01 ± 0.02	0.01	<0.01 ± 0.02	<0.01
Gymnothorax moringa	CAR	0.03 ± 0.20	0.02	0.12 ± 0.39	0.09	0.07 ± 0.30	0.05	0.06 ± 0.28	0.04	0.02 ± 0.17	0.02	0.04 ± 0.23	0.03
Gymnothorax vicinus	CAR	–	–	0.07 ± 0.30	0.06	0.03 ± 0.20	0.02	–	–	0.09 ± 0.68	0.02	0.05 ± 0.50	0.01
Labrisomus nuchipinnis	CAR	–	–	–	–	–	–	0.14 ± 0.71	0.08	0.04 ± 0.22	0.03	0.08 ± 0.51	0.05
Lutjanus alexandrei	CAR	–	–	–	–	–	–	–	–	0.05 ± 0.54	0.01	0.03 ± 0.39	<0.01
Lutjanus analis	CAR	2.94 ± 12.57	0.07	–	–	1.67 ± 9.54	0.04	–	–	–	–	–	–
Lutjanus jocu	CAR	–	–	3.87 ± 9.26	0.17	1.67 ± 6.33	0.07	–	–	–	–	–	–

(*continued on next page*)

Appendix I (*continued*)

Species	Guild[a]	Artificial reefs						Natural reefs					
		Victory ($n = 46$)		Bellucia ($n = 35$)		Total		Escalvada ($n = 113$)		Rasas ($n = 126$)		Total	
		Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O	Biomass (g m^{-2})	F_O
Lutjanus synagris	CAR	0.29 ± 1.99	0.02	–	–	0.17 ± 1.50	0.01	–	–	–	–	–	–
Malacanthus plumieri	CAR	–	–	–	–	–	–	0.08 ± 0.41	0.04	0.17 ± 0.81	0.07	0.13 ± 0.65	0.06
Ocyurus chrysurus	CAR	0.03 ± 0.18	0.07	–	–	0.02 ± 0.13	0.04	0.36 ± 3.03	0.04	0.08 ± 0.67	0.02	0.21 ± 2.14	0.03
Odontoscion dentex	CAR	–	–	8.43 ± 16.44	0.40	3.64 ± 11.51	0.17	0.41 ± 2.52	0.04	0.22 ± 1.30	0.04	0.31 ± 1.97	0.04
Ogcocephalus vespertilio	CAR	–	–	0.42 ± 1.74	0.06	0.18 ± 1.15	0.02	–	–	–	–	–	–
Pagrus pagrus	CAR	–	–	–	–	–	–	1.47 ± 11.11	0.05	0.05 ± 0.52	0.01	0.72 ± 7.66	0.03
Paralichthys brasiliensis	CAR	–	–	–	–	–	–	0.73 ± 5.75	0.02	–	–	0.35 ± 3.96	0.01
Rypticus saponaceus	CAR	0.68 ± 2.71	0.11	–	–	0.39 ± 2.06	0.06	0.19 ± 1.59	0.04	0.38 ± 1.85	0.08	0.29 ± 1.73	0.06
Scorpaena brasiliensis	CAR	–	–	–	–	–	–	0.10 ± 0.76	0.04	–	–	0.05 ± 0.53	0.02
Scorpaena isthmensis	CAR	–	–	–	–	–	–	0.01 ± 0.16	0.01	–	–	0.01 ± 0.11	<0.01
Scorpaena plumieri	CAR	–	–	1.32 ± 4.24	0.11	0.57 ± 2.84	0.05	0.39 ± 2.32	0.05	0.03 ± 0.25	0.02	0.20 ± 1.61	0.03
Aulostomus strigosus	PIS	–	–	1.12 ± 4.78	0.06	0.48 ± 3.17	0.02	1.48 ± 6.06	0.14	0.40 ± 2.04	0.06	0.91 ± 4.44	0.10
Carangoides ruber	PIS	–	–	–	–	–	–	–	–	0.01 ± 0.06	0.01	<0.01 ± 0.04	<0.01
Caranx crysos	PIS	26.08 ± 66.11	0.35	–	–	14.81 ± 51.26	0.20	28.89 ± 145.60	0.13	5.65 ± 56.14	0.02	16.64 ± 108.48	0.07
Caranx hippos	PIS	–	–	–	–	–	–	–	–	0.12 ± 1.37	0.01	0.06 ± 0.99	<0.01
Caranx latus	PIS	–	–	–	–	–	–	0.12 ± 1.29	0.01	–	–	0.06 ± 0.89	<0.01
Mycteroperca acutirostris	PIS	5.19 ± 12.50	0.20	0.96 ± 5.68	0.03	3.36 ± 10.30	0.12	–	–	0.12 ± 1.40	0.01	0.07 ± 1.01	<0.01
Mycteroperca bonaci	PIS	1.28 ± 5.14	0.07	0.44 ± 2.59	0.03	0.92 ± 4.23	0.05	0.63 ± 3.98	0.03	–	–	0.30 ± 2.75	0.01
Mycteroperca interstitialis	PIS	5.20 ± 17.68	0.20	–	–	2.96 ± 13.51	0.11	–	–	–	–	–	–
Synodus intermedius	PIS	0.53 ± 1.80	0.26	0.05 ± 0.20	0.09	0.32 ± 1.37	0.19	0.06 ± 0.38	0.06	0.08 ± 0.44	0.05	0.07 ± 0.42	0.05
Synodus synodus	PIS	–	–	0.69 ± 3.49	0.09	0.30 ± 2.30	0.04	0.04 ± 0.31	0.04	0.08 ± 0.49	0.03	0.06 ± 0.42	0.03
Gran total		**372.53 ± 467.69**		**1531.67 ± 1729.89**		**873.40 ± 1314.80**		**295.48 ± 304.34**		**226.68 ± 226.15**		**259.21 ± 267.65**	

[a] Guilds acronyms: ROV = roving herbivores, TEH = territorial herbivores, MIF = mobile invertebrate feeders, SIF = sessile invertebrate feeders, OMN = omnivores, PLK = planktivores, CAR = carnivores and PIS = piscivores.

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